

FSUI0365

## Indole

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 吲哚  
Indole

**Cat No. :** I/0365/46  
**Synonyms** 2,3-Benzopyrrole  
**CAS No** 120-72-9  
**Molecular Formula** C8 H7 N

**Supplier** **UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

**Emergency Telephone Number** Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Solid

**Appearance**  
Off-white

**Odor**  
Characteristic

#### Emergency Overview

Harmful if swallowed. Toxic in contact with skin. May cause an allergic skin reaction. Causes serious eye damage. Sensitivity to light.

#### Classification of the substance or mixture

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 4 |
| Acute Dermal Toxicity             | Category 3 |
| Serious Eye Damage/Eye Irritation | Category 1 |
| Skin Sensitization                | Category 1 |

#### Label Elements

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## Signal Word

Danger

## Hazard Statements

H302 - Harmful if swallowed  
H311 - Toxic in contact with skin  
H317 - May cause an allergic skin reaction  
H318 - Causes serious eye damage

## Precautionary Statements

## Prevention

P270 - Do not eat, drink or smoke when using this product  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P330 - Rinse mouth  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse

## Storage

P405 - Store locked up

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Health Hazards

Harmful if swallowed. Toxic in contact with skin. May cause an allergic skin reaction. Causes serious eye damage.

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

## Other Hazards

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No   | Weight % |
|-----------|----------|----------|
| Indole    | 120-72-9 | >95      |

## SECTION 4. FIRST AID MEASURES

## General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

## Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

## Skin Contact

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Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

Causes severe eye damage. May cause allergic skin reaction. RAPIDLY ABSORBED THROUGH SKIN: May be harmful if absorbed through skin: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid dust formation. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

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**Storage**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from direct sunlight.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments                                                                 |
|----------------|-------------------|-----------------|-------------|--------------------------------------------------------------------------------|
| Nitrile rubber | > 480 minutes     | > 0.1 mm        | EN 374      | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use**

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water

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system. Local authorities should be advised if significant spillages cannot be contained.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         |                                 |                                   |
|-----------------------------------------|---------------------------------|-----------------------------------|
| Appearance                              | Off-white                       |                                   |
| Physical State                          | Solid                           |                                   |
| Odor                                    | Characteristic                  |                                   |
| Odor Threshold                          | No data available               |                                   |
| pH                                      | 5.9                             | 1000g/L (20°C)                    |
| Melting Point/Range                     | 51 - 53 °C / 123.8 - 127.4 °F   |                                   |
| Softening Point                         | No data available               |                                   |
| Boiling Point/Range                     | 253 - 254 °C / 487.4 - 489.2 °F | @ 760 mmHg                        |
| Flash Point                             | 136 °C / 276.8 °F               | Method - No information available |
| Evaporation Rate                        | Not applicable                  | Solid                             |
| Flammability (solid,gas)                | No information available        |                                   |
| Explosion Limits                        | No data available               |                                   |
| Vapor Pressure                          | 0.012 mmHg @ 25 °C              |                                   |
| Vapor Density                           | Not applicable                  | Solid                             |
| Specific Gravity / Density              | 1.220                           |                                   |
| Bulk Density                            | No data available               |                                   |
| Water Solubility                        | 2.80 g/L (25°C)                 |                                   |
| Solubility in other solvents            | No information available        |                                   |
| Partition Coefficient (n-octanol/water) |                                 |                                   |
| Component                               | log Pow                         |                                   |
| Indole                                  | 2.14                            |                                   |
| Autoignition Temperature                | No data available               |                                   |
| Decomposition Temperature               | > 253°C                         |                                   |
| Viscosity                               | Not applicable                  | Solid                             |
| Explosive Properties                    | No information available        |                                   |
| Oxidizing Properties                    | No information available        |                                   |
| Molecular Formula                       | C8 H7 N                         |                                   |
| Molecular Weight                        | 117.15                          |                                   |

## SECTION 10. STABILITY AND REACTIVITY

|                          |                                                                              |
|--------------------------|------------------------------------------------------------------------------|
| Stability                | Sensitivity to light.                                                        |
| Hazardous Reactions      | None under normal processing.                                                |
| Hazardous Polymerization | Hazardous polymerization does not occur.                                     |
| Conditions to Avoid      | Incompatible products. Excess heat. Avoid dust formation. Exposure to light. |
| Materials to avoid       | Strong oxidizing agents.                                                     |

**Hazardous Decomposition Products** Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component | LD50 Oral             | LD50 Dermal                 | LC50 Inhalation |
|-----------|-----------------------|-----------------------------|-----------------|
| Indole    | LD50 = 1 g/kg ( Rat ) | LD50 = 790 mg/kg ( Rabbit ) |                 |

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|                                            |                                                                                                                                                                                                                                                               |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (b) skin corrosion/irritation;             | No data available                                                                                                                                                                                                                                             |
| (c) serious eye damage/irritation;         | Category 1                                                                                                                                                                                                                                                    |
| (d) respiratory or skin sensitization;     |                                                                                                                                                                                                                                                               |
| Respiratory                                | No data available                                                                                                                                                                                                                                             |
| Skin                                       | Category 1                                                                                                                                                                                                                                                    |
|                                            | May cause sensitization by skin contact                                                                                                                                                                                                                       |
| (e) germ cell mutagenicity;                | No data available                                                                                                                                                                                                                                             |
| (f) carcinogenicity;                       | No data available                                                                                                                                                                                                                                             |
|                                            | There are no known carcinogenic chemicals in this product                                                                                                                                                                                                     |
| (g) reproductive toxicity;                 | No data available                                                                                                                                                                                                                                             |
| (h) STOT-single exposure;                  | No data available                                                                                                                                                                                                                                             |
| (i) STOT-repeated exposure;                | No data available                                                                                                                                                                                                                                             |
| Target Organs                              | No information available.                                                                                                                                                                                                                                     |
| (j) aspiration hazard;                     | Not applicable                                                                                                                                                                                                                                                |
|                                            | Solid                                                                                                                                                                                                                                                         |
| Symptoms / effects, both acute and delayed | RAPIDLY ABSORBED THROUGH SKIN: May be harmful if absorbed through skin: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Fish                           | Water Flea | Freshwater Algae | Microtox |
|-----------|-------------------------------------------|------------|------------------|----------|
| Indole    | Onchorhynchus mykiss:<br>LC50=10 mg/L 24h | 1 mg/L 48h |                  |          |

**Persistence and Degradability** Readily biodegradable  
**Persistence** Persistence is unlikely.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Indole    | 2.14    | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

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|                                        |                                                                           |
|----------------------------------------|---------------------------------------------------------------------------|
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance            |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance            |

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Should not be released into the environment. Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations. |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point.                                                                                                                                           |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.                |

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

|                                |                              |
|--------------------------------|------------------------------|
| <b>UN-No</b>                   | UN2811                       |
| <b>Proper Shipping Name</b>    | Toxic solid, organic, n.o.s. |
| <b>Technical Shipping Name</b> | Indole                       |
| <b>Hazard Class</b>            | 6.1                          |
| <b>Packing Group</b>           | III                          |

IMDG/IMO

|                                |                              |
|--------------------------------|------------------------------|
| <b>UN-No</b>                   | UN2811                       |
| <b>Proper Shipping Name</b>    | Toxic solid, organic, n.o.s. |
| <b>Technical Shipping Name</b> | Indole                       |
| <b>Hazard Class</b>            | 6.1                          |
| <b>Packing Group</b>           | III                          |

IATA

|                                |                               |
|--------------------------------|-------------------------------|
| <b>UN-No</b>                   | UN2811                        |
| <b>Proper Shipping Name</b>    | TOXIC SOLID, ORGANIC, N.O.S.* |
| <b>Technical Shipping Name</b> | Indole                        |
| <b>Hazard Class</b>            | 6.1                           |
| <b>Packing Group</b>           | III                           |

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>Special Precautions for User</b> | No special precautions required |
|-------------------------------------|---------------------------------|

## SECTION 15. REGULATORY INFORMATION

**International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Indole    | -                                                   | X                                       | X    | X     | 204-420-7 | X    | X   | X     | X    | X    | X    | KE-21007 |

## Indole

## National Regulations

## SECTION 16. OTHER INFORMATION

**Creation Date** 21-Sep-2010  
**Revision Date** 04-Apr-2024  
**Revision Summary** Not applicable.

**Training Advice**  
Chemical incident response training.

Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**